Modeling Early Learning Gains from Sesame Street: An Evaluation of 1970s Preschool Test Scores

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1. INTRODUCTION

Promoting early childhood learning should be a key priority in ensuring the quality of education for families across the United States. Designed primarily for preschool-aged children, Sesame Street is one of the most widely studied early education television programs that focuses on foundational learning skills like early literacy and numeracy. This study analyzes trends between Sesame Street viewing habits, socioeconomic backgrounds, and pre- and post-test performance of children in the U.S. aged 3-5 to explore whether exposure to educational television is associated with significant improvements in early education, and whether these effects differ across socioeconomic groups. Understanding these trends is not only important to families who hope to supplement their child's learning, but is also important to policymakers, media producers, and teachers in informing effective methods to design television series and curriculum or learning strategies to best support early education and potentially reduce early learning gaps caused by socioeconomic differences. In the long run, the implications of this type of study can serve to improve the education of the next generation.

Our goals are to (i) evaluate which factors (e.g. age, socioeconomic status, and viewing frequency) are most influential in improving test scores, (ii) understand how socioeconomic status impacts preschool learning improvement and make predictions about these impacts, (iii) extend our methods to better understand the role of educational television in early childhood education using modern day educational examinations, and (iv) to apply our findings to inform future educational resource and curriculum design. We hypothesize that higher socioeconomic status and higher viewing frequency of Sesame Street are associated with greater increases in various tested areas. To achieve our goals, we analyze and create models from a dataset with a variety of variables for different pre- and post-viewing test scores of preschool children.

One limitation of this dataset is that it is not particularly modern (i.e. early 1970s), and there is an innately large amount of noise due to not fully capturing outside developmental influences confounding test results. These limitations lead us to prioritize drawing insightful inferences over making the most accurate prediction as a project goal. The specific tests in the dataset are not synonymous to those used today, and the inherently noisy dataset involving children’s cognitive development will make exact prediction difficult and less useful than drawing general insights about influential factors.

Our objectives allow us to assess both the overall effectiveness of Sesame Street and how its impact may differ across populations. Our intended audiences benefit from both the statistical findings and the practical insights of our study. We hope that findings from this study are used in the future to better serve education in underserved communities, supporting efforts to improve education accessibility and equity, and overall, we hope this project contributes to understanding how accessible educational resources can complement formal preschool education.

2. METHODOLOGY

2.1 Data & Exploratory Data Analysis

This dataset comes from an early 1970s Educational Testing Service study evaluating the impact of *Sesame Street* on preschool children from diverse socioeconomic backgrounds across five U.S. sites. It includes pre- and post-test measures of cognitive skills, viewing behavior, and an experimental encouragement condition designed to assess whether watching *Sesame Street* improved learning outcomes and helped economically disadvantaged children catch up to their more advantaged peers. The original dataset, which had 240 observations, did not have any missing values to address. Additionally, due to the highly noisy nature of the 1970s data and the fact that these outdated tests are not directly synonymous to current-day developmental tests for preschoolers, we elected not to use an additional dataset with modern preschool tests; joining more datasets would not only add noise and complexity to the data and models, but incompatibility and transferability between old and new test scores as well as the new improvement variable we created may cause further issues like extrapolation (i.e. child-to-child) or drawing difficult-to-interpret inferences (i.e. child-to-group). Since our goal is not solely to accurately predict the learning outcome of individual children, but more so to draw insightful inferences to inform policy and curriculum to improve preschool education, we chose to focus on the original dataset.

Variables of Interest

Pre- and Post-tests in the following areas:

- `body`: body parts (scores range from 0-32)
- `let`: letters (scores range from 0-58)
- `form`: recognizing and naming shapes (scores range from 0-20)
- `numb`: numbers (scores range from 0-54)
- `relat`: amount, size, and position relationships, e.g. larger/smaller, first/last (scores range from 0-17)
- `clasf`: classification skills (scores range from 0-24)

Since each test had a different range of scores, we standardized all pre- and post-test scores to be reported as a percentage of the total possible score (`post_sesame_percentage - pre_sesame_percentage`). These are our **Response Variables**.

Covariates:

- `sex`: male = 1, female = 2
- `age`: age in months
- `peabody`: mental age score obtained from administration of the Peabody Picture Vocabulary test as a pretest measure of vocabulary maturity
- `setting`: setting in which Sesame Street was viewed, 1=home 2=school

Primary Predictor Variables:

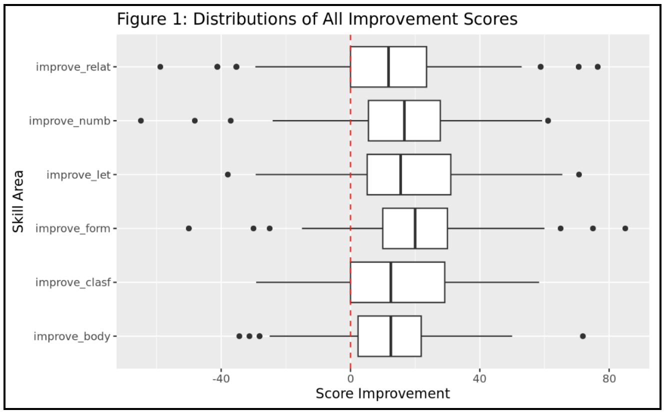
- `viewenc`: treatment condition (child encouraged to watch, child not encouraged to watch)
- `viewcat`: frequency of viewing, (rarely watched the show, once or twice a week, three to five times a week, watched the show on average more than 5 times a week)
- `site`: (Three to five year old disadvantaged children from inner city areas in various parts of the country, Four year old advantaged suburban children, Advantaged rural children, Disadvantaged rural children, Disadvantaged Spanish speaking children)

Table 1: Mean Improvement by Viewing Category

viewcat	body	letter	form	number	relation	classify
1	9.90	4.28	13.89	8.50	6.86	8.72
2	11.82	14.40	18.17	15.65	9.02	13.82
3	12.21	25.86	19.45	20.20	9.56	16.08
4	14.01	27.75	24.03	22.22	15.09	18.95

The mean improvement by viewing category table shows a **consistently strong positive relationship between viewing frequency and improvement across all test areas**, suggesting that Sesame Street exposure is associated with larger learning improvements. As viewing frequency increased from 1 to 4, **letter test scores showed the greatest increase** in percent improvement on average (from 4.28% to 27.75%).

The largest average percent improvements occurred in the tests for forms (19.06%), letters (18.63%), and numbers (16.95%), indicating the **strongest learning growth in forms, letters and numbers after Sesame Street exposure**. In contrast, there were only modest improvements in body knowledge (12.06%) and classification skills (14.60%) test scores and weak improvement in relational terms test scores (10.25%). All tests had large negative minimums (as low as -64.81%) and high maximums (all at least 58.33%), indicating that there was greater variability in the improvements or lack thereof in these tested areas for various students.



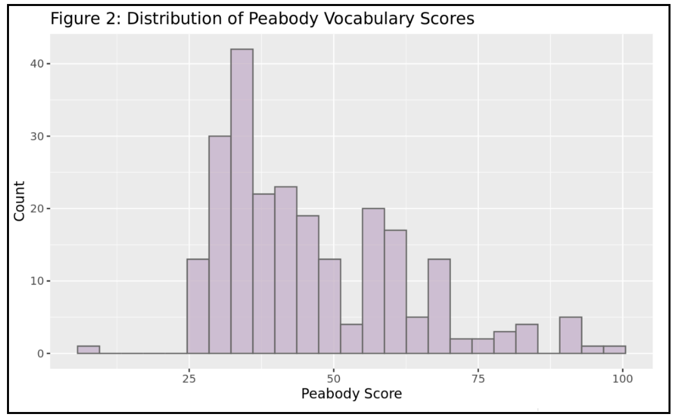
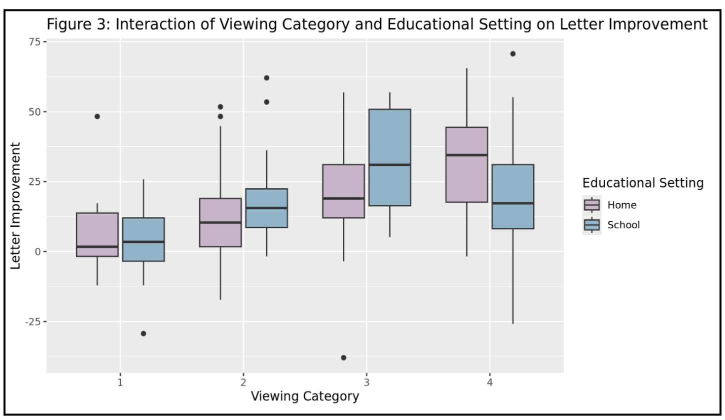


Figure 1 displays that the distributions of percent improvement across the 6 test areas showed **positive median improvement percentages for all 6 test areas**, with the largest median improvements observed for the tests on **forms, numbers, and letters** and the greatest variability for **forms, numbers, and relational terms**. Most children demonstrated learning gains, but various outliers across almost all tests indicate variability in children's responses to Sesame Street exposure. Figure 2 displays that the distribution of Peabody scores is **right-skewed**, indicating that most children are clustered in the lower score range.



Letter test score improvement increases consistently with viewing frequency for viewing at home (setting=1), but for viewing at a school setting (setting=2), letter score test percent improvement slightly decreased at the highest viewing frequency. At **low-to-moderate viewing categories, viewing at school resulted in greater median letter test score improvements**, while at the highest viewing frequency, viewing at school resulted in a lower improvement than viewing at home; this could suggest that **too high of a viewing frequency in a school setting is actually detrimental to early learning**. Lastly, variability generally increases with higher viewing categories, showing that greater levels of Sesame Street exposure can lead to a larger variety of test outcomes for children.

Overall, the EDA shows that **Sesame Street exposure is positively associated with learning gains**, with the **largest improvements in forms, letters, and numbers**. **Viewing frequency displayed a positive relationship** with higher exposure linked to larger average improvements, especially for letters, though gains varied by setting at the highest viewing level. **All outcomes showed substantial variability and notable negative outliers** (retained during modeling), indicating that while most children benefited, responses differed widely across individuals.

2.2 Model Fitting

To understand how Sesame street exposure and socioeconomic context relate to test score improvements, we fit six separate models, one for each outcome (body parts, letters, forms, numbers, relational terms, and classification), all using the model-fitting procedure below.

2.2.1 Transformations

Response: As mentioned in EDA, for each test, we first converted the raw pre- and post-test scores to percentages of the test's maximum score to account for different score ranges across tests. We then defined an improvement measure as the difference between the post-test percentage and pre-test percentage (e.g. `improve_body = postbody_p - prebody_p`), so that positive values reflect learning gains and negative values reflect declines. These improvement measures lie in the interval [-100,100] (percentages). To fit beta regression models, we rescaled each variable to the open interval (0,1) by dividing by 100 to obtain values in [-1,1], and then applying the following linear transformation to yield the six responses used for modeling.

$$Betatransf = (improvement/100 + 1)/2$$

We chose a continuous regression approach rather than treating outcomes as ordered categorical variables because converting test score improvements into categories comparable between tests would require arbitrary binning. This would result in model performance and insights entirely dependent on the number of binned categories, worsening granularity. Doing so would also render the last 20% of the data unusable if we were to use sufficiently few bins (e.g. 5) to train our models for once-versus-one classification. Furthermore, as the number of levels in an ordered categorical variable increases, that variable increasingly resembles a continuous outcome, thereby justifying modeling improvement as a proportion.

Predictors: `peabody` was log-transformed in order to reduce skewness and stabilize relationships. The `site` indicator, capturing both socioeconomic and geographic context, was reordered so that its ordinal factor levels would reflect an approximate socioeconomic gradient: disadvantaged Spanish-speaking children, disadvantaged inner-city children, disadvantaged rural children, advantaged rural children, and advantaged suburban children. `viewcat` was converted to an ordinal variable following its initial ordering. We also renamed `sex`, `setting`, and `viewenc` to have meaningful factor names. These transformations allow for the model to capture potentially nonlinear effects of baseline vocabulary (via the logarithm), preserve the ordered nature of viewing frequency and site categories, and allow for interpretable predictor labels. Lastly, we standardized `peabody_log` and `age` to have a mean of 0 and a standard deviation of 1.

2.2.2 Model choice: Beta regression with interactions and Lasso regularization

For each rescaled improvement outcome, we fit a separate **beta regression** model with a logit link function using the `bamlss` package in R. Each response is a continuous proportion between 0 and 1, so beta regression is a natural choice. This is preferable to a standard linear regression model since the latter ignores these bounds, allows fitted values outside the feasible range, and relies on a normal error structure that is not appropriate for proportions. We also considered censored normal (Tobit) models as an alternative, but these treat the outcome as approximately Gaussian rather than directly modeling proportions, so we ultimately favored beta regression.

In all six models, we relate the improvement to the predictors using a logit link function:

$$logit(improvement\ for\ test\ k\ and\ child\ i) = linear\ predictor\ for\ test\ k\ and\ child\ i,$$

Exploratory data analysis revealed that several relationships are not purely additive. For example, the association between viewing frequency (`viewcat`) and improvement in letter scores appeared to depend on whether children watched at home or at school (`setting`), and patterns varied across sites and levels of baseline vocabulary. To capture this type of effect modification while keeping the model interpretable, we started with a specification including: i) all main effects of sex, age, peabody_log, setting, viewenc, viewcat, and site, and ii) all their pairwise interaction terms.

This full two-way interaction structure is high-dimensional, especially because several predictors are multi-level factors. To prevent overfitting and to focus on the most influential effects, we used lasso regularization via the `la()` term specification in `bamlss`, with `type="single"` to impose a single penalty parameter λ across all terms. We fitted this using the `opt_lasso` optimizer over a grid of 100 candidate penalty values. Then, we selected the penalty level by minimizing the BIC, which penalizes model complexity and is consistent with our primary interest in inference rather than pure predictive accuracy. Our prioritization of inference is also what led us to choose lasso regularization and simple transformations rather than splines: while the usage of other non-linear modeling may have been better suited to fit our data, we opted for simple lasso instead in order to ensure interpretability in the presence of interaction terms.

2.2.3 Variable Selection

For each of the six outcomes, we first fit this beta-lasso model with all main effects and pairwise interactions. In our setting, however, the `opt_lasso` optimizer in `bamlss` rarely drives coefficients all the way to zero. Instead, strongly penalized terms end up with very small, ridge-like estimates rather than being removed outright. Additionally, because beta regression provides distributional estimates for coefficients, it is rare for the actual coefficient estimate (the distribution mean) to equal exactly zero. For this reason, we treat this penalized fit as a screening step rather than a complete selection procedure. A central part of our methodology is a second, explicit variable selection stage built on top of the lasso fit, in which we systemically decide which terms to retain or discard using the provided 95% confidence intervals for each coefficient**.** If the 95% interval for a coefficient contained zero, we treated that term as not clearly different from zero, and thus dropped the term.

Figure 4: BIC vs. log(lambda) For The Letters model

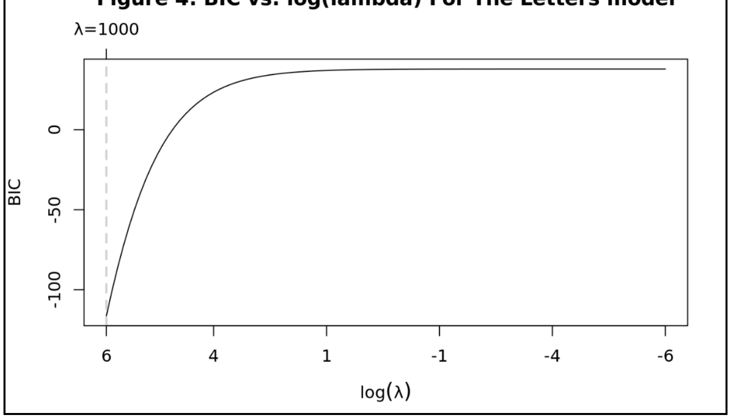


Figure 4 displays how BIC was used to select the Lasso tuning parameter λ for each model, using Letters as an example. The displayed BIC curve indicates that a relatively large value of λ minimized BIC, implying that stronger regularization provided the best trade-off between model fit and complexity. As a result, several weaker effects are shrunk toward zero to yield a more parsimonious model.

For categorical predictors, we only removed the entire factor if all levels had confidence intervals including zero. If at least one level showed evidence of a non-zero effect, we kept the factor in the model. We also enforced a standard hierarchy rule: when an interaction term (e.g. viewing frequency × setting) was retained, the corresponding main effects were kept even if their individual confidence intervals included zero.

Applying this procedure produced a reduced specification for each response that contained all main effects as well as interactions with confidence intervals not including zero. Henceforth, we refer to the models refitted using these reduced specifications as the adjusted models.

2.2.4 Model Comparison and Performance

Next, we compared the initial full-interaction models and the adjusted models using two criteria: the BIC and training root mean squared error (RMSE). We used BIC during fitting to select the lasso penalty level and to compare model specifications, with a lower BIC indicating a better balance between fit and complexity. We used training RMSE of the rescaled outcome to compare the predicted mean improvement to the observed transformed improvements.

The reason for using training RSME is twofold: first, as formerly mentioned, because our primary goal in this project is inference, we focused on building models that are interpretable and reasonably well-fitting on the full dataset rather than optimizing prediction error. The Sesame Street data comes from a specific cohort of children tested in the early 1970s, under a particular design and set of educational conditions. Modern preschool environments, tests, and media exposure look very different, so a model optimizing predictive performance on these data would not effectively transfer to today's children without significant extrapolation. Also, because the sample size is relatively small, aggressively partitioning the data into training and testing sets or running extensive cross-validation would further reduce the sample available for estimation.

Secondly, there is a practical consideration: the lasso penalty in our models is implemented through `la()` within `bamlss`. These terms are estimated jointly with the smoothing and penalty structure inside `bamlss`, and due to intricacies of the ways in which `la()` within `bamlss` is implemented, it is not possible to use this framework for prediction on an out-of-training sample. While this would be a considerable limitation for most modeling procedures, because we have no particular use for prediction and are instead entirely interested in general inference and conclusions, this is not an issue. In our workflow, we therefore relied on training RMSE as a descriptive measure of in-sample fit and used it alongside BIC to compare and refine specifications.

For three of the six outcomes (body, relational terms, and classification), the adjusted models achieved lower RMSE and better BIC values than the initial full-interaction models. This suggests that removing weak interactions improved both parsimony and fit to our observed data for three of the models, but regularization without full penalization was sufficient for the other three. For the letters, forms, and numbers models, the initial models had slightly better (lower) training RMSE, so we retained those full specifications as the final models for these test areas. The inference results are based on these six final models.

Table 2: Predictors Kept Across Final Models

Predictor	**body**	**letters**	**form**	**numb**	**relat**	**clasf**
age	✔	✔	✔	✔	✔	✔
age:setting	✔	✔	✔	✔	✖	✔
age:viewcat	✔	✔	✔	✔	✔	✔
age:viewenc	✔	✔	✔	✔	✔	✔
peabody_log	✔	✔	✔	✔	✔	✔
setting	✔	✔	✔	✔	✔	✔
setting:peabody_log	✖	✔	✔	✔	✔	✔
setting:viewenc	✔	✔	✔	✔	✔	✔
sex	✔	✔	✔	✔	✔	✔
sex:age	✖	✔	✔	✔	✖	✔
sex:peabody_log	✖	✔	✔	✔	✖	✔
sex:setting	✖	✔	✔	✔	✔	✔
sex:viewcat	✔	✔	✔	✔	✔	✔
sex:viewenc	✔	✔	✔	✔	✔	✔
site	✔	✔	✔	✔	✔	✔
site:age	✖	✔	✔	✔	✖	✔
site:peabody_log	✖	✔	✔	✔	✖	✔
site:sex	✔	✔	✔	✔	✔	✔
site:viewcat	✔	✔	✔	✔	✔	✔
site:viewenc	✔	✔	✔	✔	✔	✔
viewcat	✔	✔	✔	✔	✔	✔
viewcat:peabody_log	✔	✔	✔	✔	✖	✔
viewcat:setting	✔	✔	✔	✔	✔	✔
viewcat:viewenc	✔	✔	✔	✔	✔	✔
viewenc	✔	✔	✔	✔	✔	✔
viewenc:peabody_log	✔	✔	✔	✔	✔	✔

<u>RESULTS</u>

3.1 Final Selected Models

Table 2 summarizes the predictors kept in each of the six final beta regression models following the lasso-based semi-manual variable-selection procedure discussed in Methodology. Across all outcomes, the main effects of `age`, `site`, viewing frequency (`viewcat`), viewing `setting`, viewing encouragement (`viewenc`), and baseline vocabulary score (`peabody_log`) were consistently retained, indicating that children's learning gains are influenced by a combination of developmental factors, socioeconomic context, and exposure to Sesame Street. Below is an example of the full model specification, which would be altered for each of the six models, where '...' denotes additional interaction terms as detailed in Table 2, '*' denotes a standardized variable, and `logit_var` indicates the logit transformation of the improvement percentage scores for each test.

$$logitvar = B_0 + B_1 age* + B_2(log(peabody)*) + B_3 setting + B_4 sex + B_5 site + B_6 viewcat + B_7 viewenc + ...$$

Table 3 displays the training root mean squared errors for each final model. Training RMSE is provided as opposed to test RMSE, as is briefly validated in Methodology. We observe that the lowest RMSE belongs to the Body and Numbers models, while the largest RMSE belongs to the Forms model. On average, the Body and Numbers model provides estimates for improvement score with residuals of 0.138 units while the Forms model provides estimates with residuals of 0.257.

Table 3: Training RMSE for Final Models

Model	RMSE
Body	0.138
Letters	0.178
Forms	0.257
Numbers	0.138
Relational	0.192
Classification	0.205

3.2 Interpretation

Table 4: Top 3 Most Influential Predictors in Each Final Model

model	term	estimate	abs_est
body	mu.s.la(~sex:viewcat).sexmale:viewcatonce_twice	1.2645	1.2645
body	mu.s.la(~sex:viewcat).sexfemale:viewcatonce_twice	1.2390	1.2390
body	mu.s.la(site).siteadv_suburb	-1.1253	1.1253
letters	mu.s.la(viewcat).viewcatthree_five	-1.9528	1.9528
letters	mu.s.la(peabody_log).peabody_log	1.3408	1.3408
letters	mu.s.la(~sex:age).sexfemale:age	-1.1416	1.1416
form	mu.s.la(~sex:setting).sexmale:settinghome	-1.4257	1.4257
form	mu.s.la(~site:viewenc).sitedisadv_span:viewencencouraged	-1.4038	1.4038
form	mu.s.la(~site:viewcat).sitedisadv_span:viewcatrare	1.3153	1.3153
numb	mu.s.la(viewcat).viewcatthree_five	-1.4536	1.4536
numb	mu.s.la(~setting:peabody_log).settingschool:peabody_log	1.0800	1.0800
numb	mu.s.la(~setting:peabody_log).settinghome:peabody_log	-1.0558	1.0558
relat	mu.s.la(~setting:peabody_log).settingschool:peabody_log	1.1832	1.1832
relat	mu.s.la(~setting:peabody_log).settinghome:peabody_log	-1.1814	1.1814
relat	mu.s.la(site).sitedisadv_rural	-1.1400	1.1400
clasf	mu.s.la(viewcat).viewcatover_five	-1.9921	1.9921
clasf	mu.s.la(~sex:setting).sexfemale:settinghome	1.7770	1.7770
clasf	mu.s.la(~site:viewenc).sitedisadv_span:viewencencouraged	-1.3315	1.3315

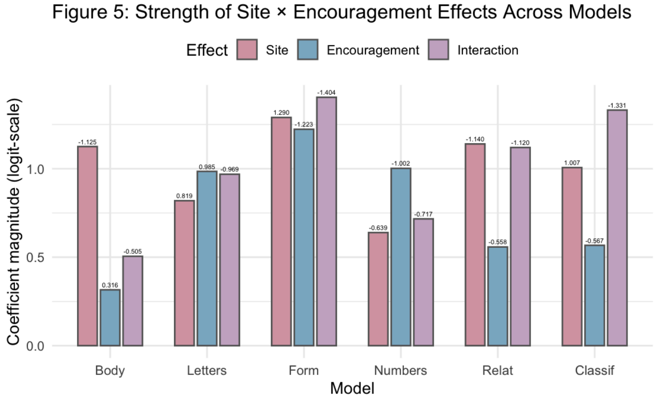
To summarize which predictors had the strongest influence in each outcome, we extracted the *mu* coefficients from our final models and ranked them by magnitude. Table 4 identifies the top three predictors for every model, highlighting the strongest associations with improvement in each outcome. This provides an interpretable snapshot of the key drivers behind improvement in each learning outcome by following several steps. First, because the logit link function was used in the final models, we interpret the coefficients accordingly; secondly, in order to use the beta regression framework, we had to transform our response variable range from [-1,1] to [0,1] at the very beginning of the modeling process. Thus, we perform a back-transformation to return to our original scale. Then, we multiplied our model output (% change in total score) by the total range of the test being modeled and rounded down. An example of an interpretation for a main effect categorical variable coefficient in the Body model would thus be the following: all else held constant, a child who is living in an advantaged rural setting will, on average, have a Body improvement score 0.4659 larger than a child who is in a disadvantaged Spanish-speaking setting. In the context of the Body model (which has a score range of 0-32), this is representative of a 14 point increase. This appears to be a large effect; however, because there are so many other predictors present in the model (with a negative effect on outcome), we assume that this remains a logical conclusion. Because comparisons between models are easiest to make using percentages, we will not perform this last step for the remainder of the report.

Two main effects, socioeconomic status (`site`) and viewing frequency (`viewcat`), appear repeatedly among the most influential predictors across models. Curiously, viewing frequency appears to differ in its impact on improvements depending on its intensity. Across five of the models, a viewing frequency of greater than five times in one week consistently positively impacts improvement scores, whereas that does not hold true for lesser frequencies; viewing Sesame Street three to five times per week yields negative impacts across 6 of the models.

`site`, modeled as an ordinal variable and reordered to reflect increasing socioeconomic advantage, captures contextual differences in children's learning environments that are consistently associated with improvement. Observing main effects for site, disadvantaged city sites and advantaged suburban sites were found to negatively impact improvement scores, while both disadvantaged rural and advantaged rural sites positively impacted scores. Similar to viewing frequency, these varying results (expected negative impact of the disadvantaged rural level, but unexpected negative impact of advantaged suburban level) from main effects for `site` **support further investigation into the effects of interactions**. Notably, the EDA also suggests potential non-linearities in this relationship, with very high viewing frequencies in school settings sometimes associated with smaller average gains and increased variability in outcomes, indicating that increased exposure does not uniformly translate into larger improvements for all children. Similarly, when looking at the `setting` main effect, we observe the coefficients to be negative for the Body and Forms model, suggesting that holding all else constant, viewing Sesame Street at home may be more beneficial for those skills, while school-based viewing supports greater learning gains for skills related to letters, numbers, relations, and classification.

While one may be able to rationalize these results on their own, upon further investigation, we realized that the results of these miscoherencies can be attributed to the fact that **interaction terms greatly change the impact of these variables**. Across models, three interaction effects consistently emerged as the most influential predictors. These patterns capture the major ways children's learning responses differed:

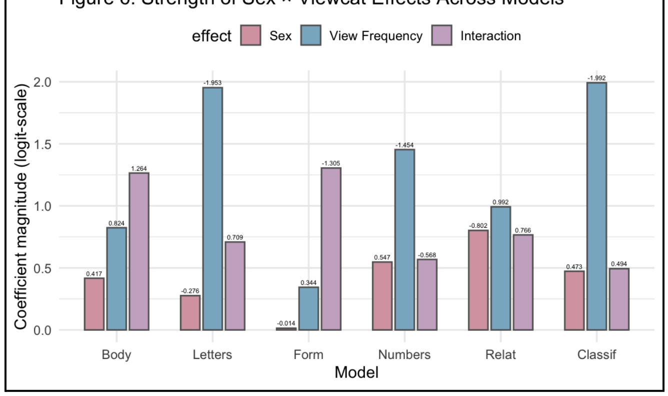
Figure 5: Strength of Site × Encouragement Effects Across Models



Interaction effects between site and encouragement are most influential for Form and Classification and less influential for Body, Letters, Numbers, and Relation in comparison to the relative main effects. Holding all else held constant, children in disadvantaged Spanish-speaking sites who are encouraged to watch Sesame Street are estimated to improve, on average, by 0.606 points less in Form and 0.582 points less in Classification than those who are not encouraged. **This finding suggests that in environments with fewer external educational resources, externally-motivated viewing can limit opportunities for children to engage with the content on their own.**

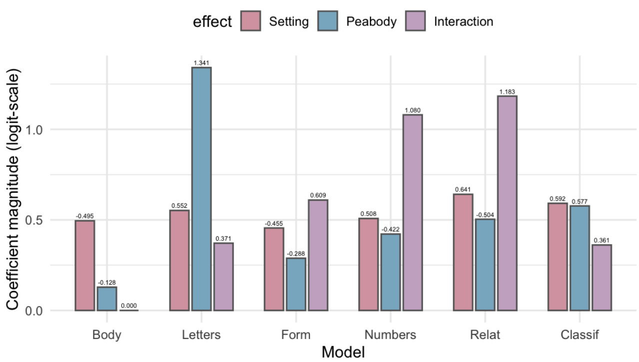
Figure 6 shows that the interactions between sex and viewing frequency are influential for Body and Form, and less influential for Letters, Numbers, Relation, and Classification, in comparison to the relative main effects. This means that, all else held constant, the association between viewing frequency and Body improvement differs by sex; for instance, among children who view Sesame Street once or twice a week, male children are estimated to have a Body improvement score 0.56 higher than female children. **This finding suggests that, at moderate viewing frequencies, male and female children may engage differently with body and movement-focused content, leading to sex-specific differences in Body improvement outcomes.**

Figure 6: Strength of Sex × Viewcat Effects Across Models



Notably, however, this interaction is not nearly as influential as the main effects for Letters, Numbers, and Classification, which suggests that there may not exist this same difference in engagement with these types of cognitive activities between sexes at different viewing frequencies.

Figure 7: Strength of Setting × Vocabulary Effects Across Models



Interaction effects between setting and vocabulary are influential for Form, Numbers, and Relation, and less influential for Body, Letters, and Classification in comparison to the relative main effects. This shows that, all else held constant, for each one-unit increase in the log-transformed and standardized Peabody vocabulary score, children viewing Sesame Street at school are predicted to experience an additional 0.532 increase in Relation improvement score, compared to children with the same vocabulary level who view Sesame Street at home. **This result suggests that school-based viewing may provide additional structure allowing children with stronger baseline language skills to more effectively translate Sesame-facilitated vocabulary growth into further relational learning outcomes.**

CONCLUSION

Our study used data from a large Sesame Street evaluation in the 1970s to examine how children's exposure to an educational television program, together with socioeconomic and developmental factors, relates to gains in early academic skills. We converted pre- and post-test scores into standardized improvement measures and fitted six separate beta regression models with interactions and lasso-based variable selection, one for each outcome area (body parts, letters, forms, numbers, relational terms, and classification). In doing so, we were able to study how age, site, viewing frequency, setting, encouragement, and baseline vocabulary jointly relate to changes in test performance rather than treating Sesame exposure in isolation.

Our EDA showed that, on average, children improved in all six areas, with the largest mean gains in forms, letters, and numbers, and more modest changes in body knowledge, classification, and relational terms. Mean improvement increased steadily across all tests, and letters displayed the largest increase from lowest to highest viewing category. At the same time, the distributions of improvement scores were wide, with substantial variability and notable negative outliers, and Peabody vocabulary scores were right-skewed, which means most children started with relatively low vocabulary levels. These patterns suggested that Sesame Street was associated with learning gains for many children, but that responses varied across individuals.

In the fitted regression models, we retained a common set of core predictors (age, site, viewing frequency, setting, encouragement, Peabody) by design and relied on a lasso and confidence interval procedure to determine which interaction terms stayed in each outcome. Training RMSE values for the final models were moderately low, with Body and Numbers achieving the lowest error and Forms the highest. The strongest estimated terms across models were predominantly interactions: combinations of site and encouragement (especially for Forms and Classifications), sex and viewing frequency (especially for Body), and setting and baseline vocabulary (for Forms, Numbers, and Relation). Together with the main effects of viewing frequency and site, these patterns indicate that **improvements were often tied to interactions where children watched Sesame Street, how they were encouraged to do so, and the socioeconomic and vocabulary context in which they viewed the program.**

Our conclusions are subject to several important limitations. As already noted earlier, the data comes from preschoolers in the early 1970s under testing instruments and educational conditions that differ from today. Many other developmental influences (e.g. home environment, parenting style, daycare or preschool) are not measured in the data, so it is inherently noisy and only useful when tied to its original context. In addition, the `site` variable is a coarse grouping that mixes geography, socioeconomic status, and age (e.g. Three to five year old disadvantaged children from inner city vs. four year old advantaged suburban children), so it is not an isolated socioeconomic status measure and should be interpreted carefully. Also, the sample size is small relative to the number of interaction terms considered, and the use of `bamlss` with `la()` terms and a semi-manual variable selection procedure means that we are focusing on in-sample RMSE and BIC rather than cross-validated prediction or external forecasting. As a result, our six final models are best interpreted as a summary of patterns in this particular evaluation rather than as a predictive tool for new children.

However, even within these constraints, our analysis offers practical takeaways for families, educators, media producers, and policymakers. Within the study, we found that **more frequent viewing is associated with larger average gains past a threshold of five viewings per week**, especially in letter-related outcomes, but the **interaction results make clear that "more is better" does not hold in the same way for every setting or group**. The role of site and its interactions with encouragement and vocabulary also highlight how **children in different socioeconomic environments do not respond identically to the same media exposure and that viewing context (home vs. school) and how viewing is encouraged or not also matter alongside mere exposure**.

Our modeling strategy (using beta regression with interactions and targeted regularization), thus, provides a template for analyzing similar educational dataset. If comparable modern data becomes available, the same approach can be used to reassess how today's educational television and digital media contribute to early learning in settings where educational content and classroom practices look very different from those of the 1970s. If this study were to be replicated in modern contexts and revealed to yield similar results, **educators and parents alike should consider encouraging frequent viewing of Sesame Street within the home and the classroom while keeping in mind that its educational impacts will differ depending on socioeconomic status, vocabulary level, and sex.**

Lastly, media producers should investigate why viewing may affect students of different socioeconomic backgrounds differently and tailor their programs to prevent this from being the case. In summary, by analyzing the effects of viewing Sesame Street, this study provides concrete recommendations drawing from the positive impacts of Sesame Street viewing among preschool-age children and demonstrates the ways in which these impacts differ across child-by-child attributes.